

R Programming

#01

Getting Started

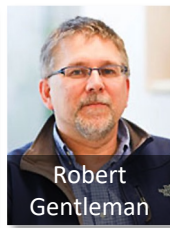
#01

INTRODUCTION TO R

R.....a mythical beast?

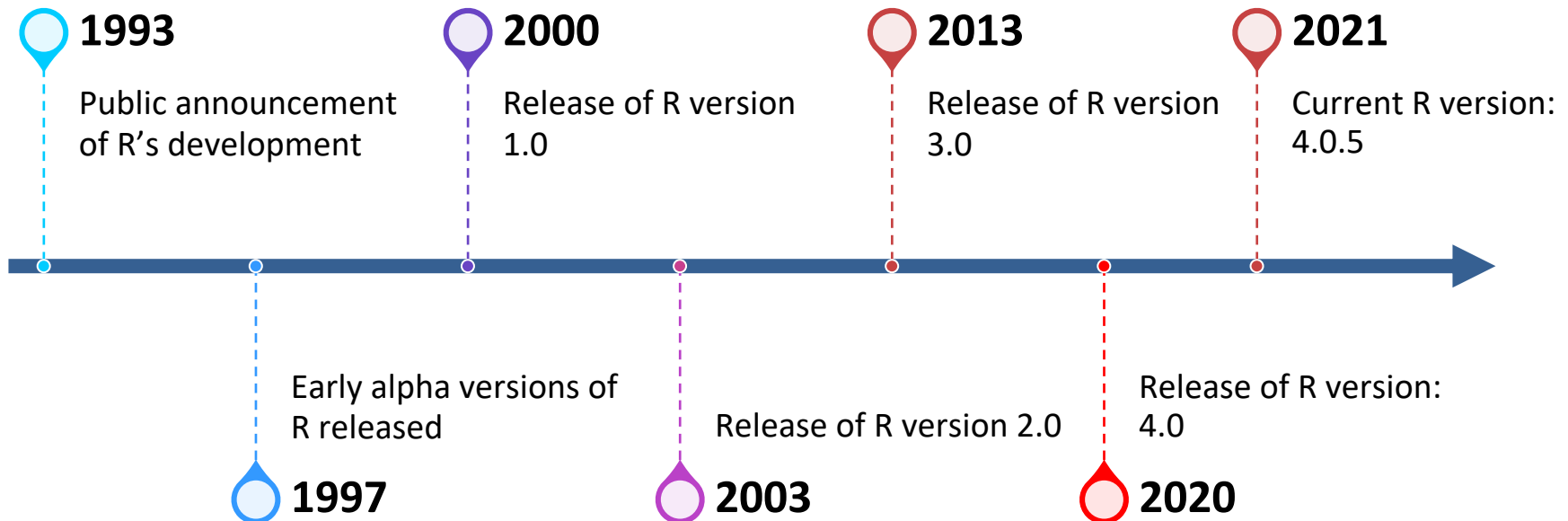


- R is **free** and **open-source**
- R is a **command-line** programming language
- It is not a statistics program! But a programming language that **works well for statistics**
- With **visualization capabilities** and **highly extensible**.
- Operating system independent (supported in: Windows / macOS and Linux)
- R can be integrated with other packages and languages (Excel, SAS, SPSS, etc.)



A Brief History

- R was developed by **Ross Ihaka** and **Robert Gentleman**
- The original goal was to develop software for students to use
- They were encouraged to make it an open-source project.

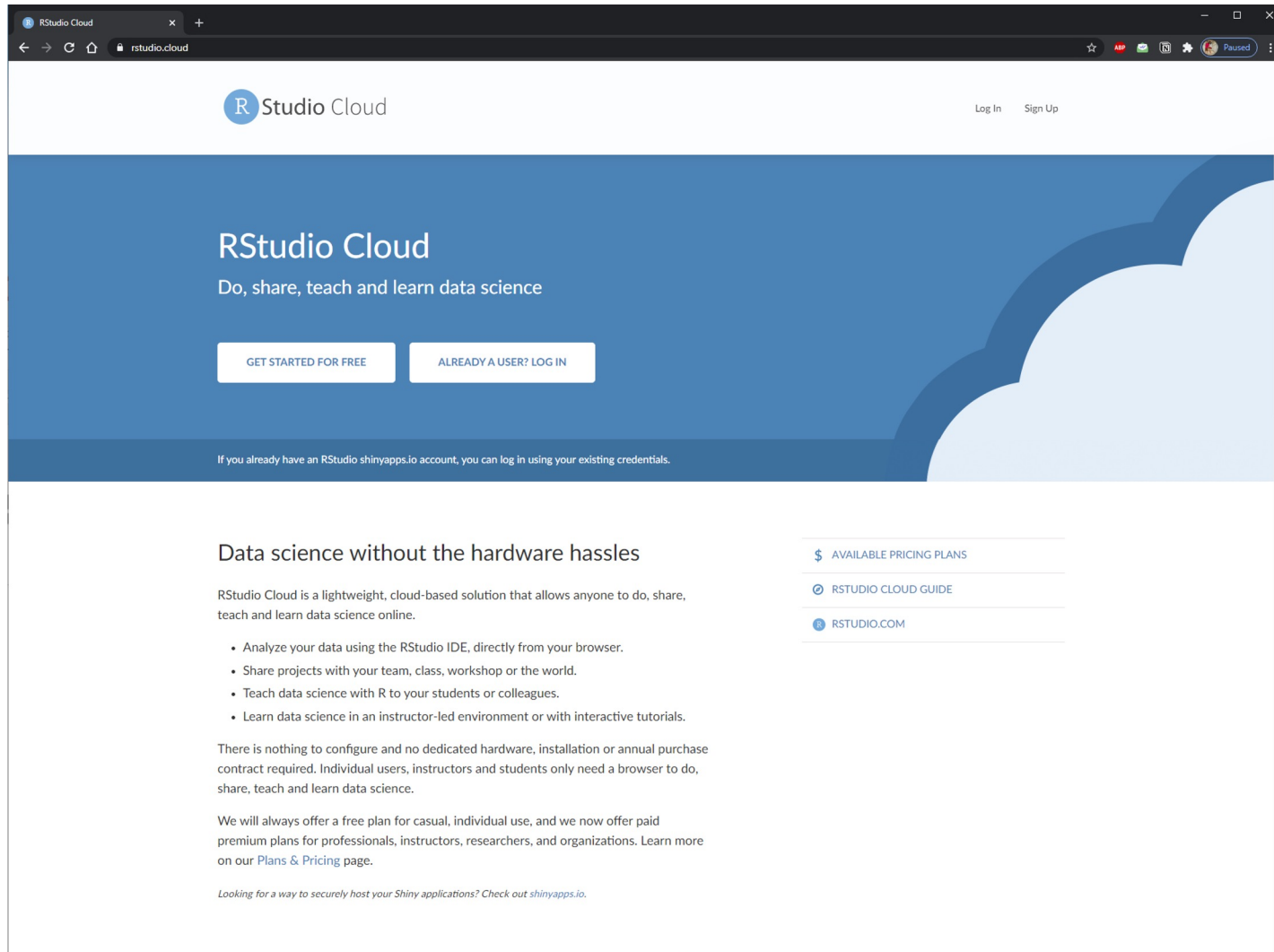


Using R

- **Two** ways:
 1. Through R environment setup in your local computer (requires installation of R environment and RStudio (optional)). **OR**
 2. Through RStudio Cloud (<https://rstudio.cloud/>)

 RStudio Cloud

Using RStudio Cloud



The screenshot shows the RStudio Cloud website in a web browser. The browser's address bar shows 'rstudio.cloud'. The website has a dark blue header with the 'R Studio Cloud' logo and 'Log In' and 'Sign Up' links. Below the header is a large blue banner with the text 'RStudio Cloud' and 'Do, share, teach and learn data science'. Two white buttons are present: 'GET STARTED FOR FREE' and 'ALREADY A USER? LOG IN'. Below the banner, a small text line states: 'If you already have an RStudio shinyapps.io account, you can log in using your existing credentials.' The main content area is white and features the heading 'Data science without the hardware hassles'. Below this heading is a paragraph: 'RStudio Cloud is a lightweight, cloud-based solution that allows anyone to do, share, teach and learn data science online.' This is followed by a bulleted list of features: 'Analyze your data using the RStudio IDE, directly from your browser.', 'Share projects with your team, class, workshop or the world.', 'Teach data science with R to your students or colleagues.', and 'Learn data science in an instructor-led environment or with interactive tutorials.' Below the list is another paragraph: 'There is nothing to configure and no dedicated hardware, installation or annual purchase contract required. Individual users, instructors and students only need a browser to do, share, teach and learn data science.' This is followed by a paragraph: 'We will always offer a free plan for casual, individual use, and we now offer paid premium plans for professionals, instructors, researchers, and organizations. Learn more on our [Plans & Pricing](#) page.' At the bottom of the main content area is a small text line: 'Looking for a way to securely host your Shiny applications? Check out [shinyapps.io](#).' On the right side of the main content area, there is a vertical list of links: '\$ AVAILABLE PRICING PLANS', 'RSTUDIO CLOUD GUIDE', and 'RSTUDIO.COM'.

RStudio Cloud

Do, share, teach and learn data science

GET STARTED FOR FREE

ALREADY A USER? LOG IN

If you already have an RStudio shinyapps.io account, you can log in using your existing credentials.

Data science without the hardware hassles

RStudio Cloud is a lightweight, cloud-based solution that allows anyone to do, share, teach and learn data science online.

- Analyze your data using the RStudio IDE, directly from your browser.
- Share projects with your team, class, workshop or the world.
- Teach data science with R to your students or colleagues.
- Learn data science in an instructor-led environment or with interactive tutorials.

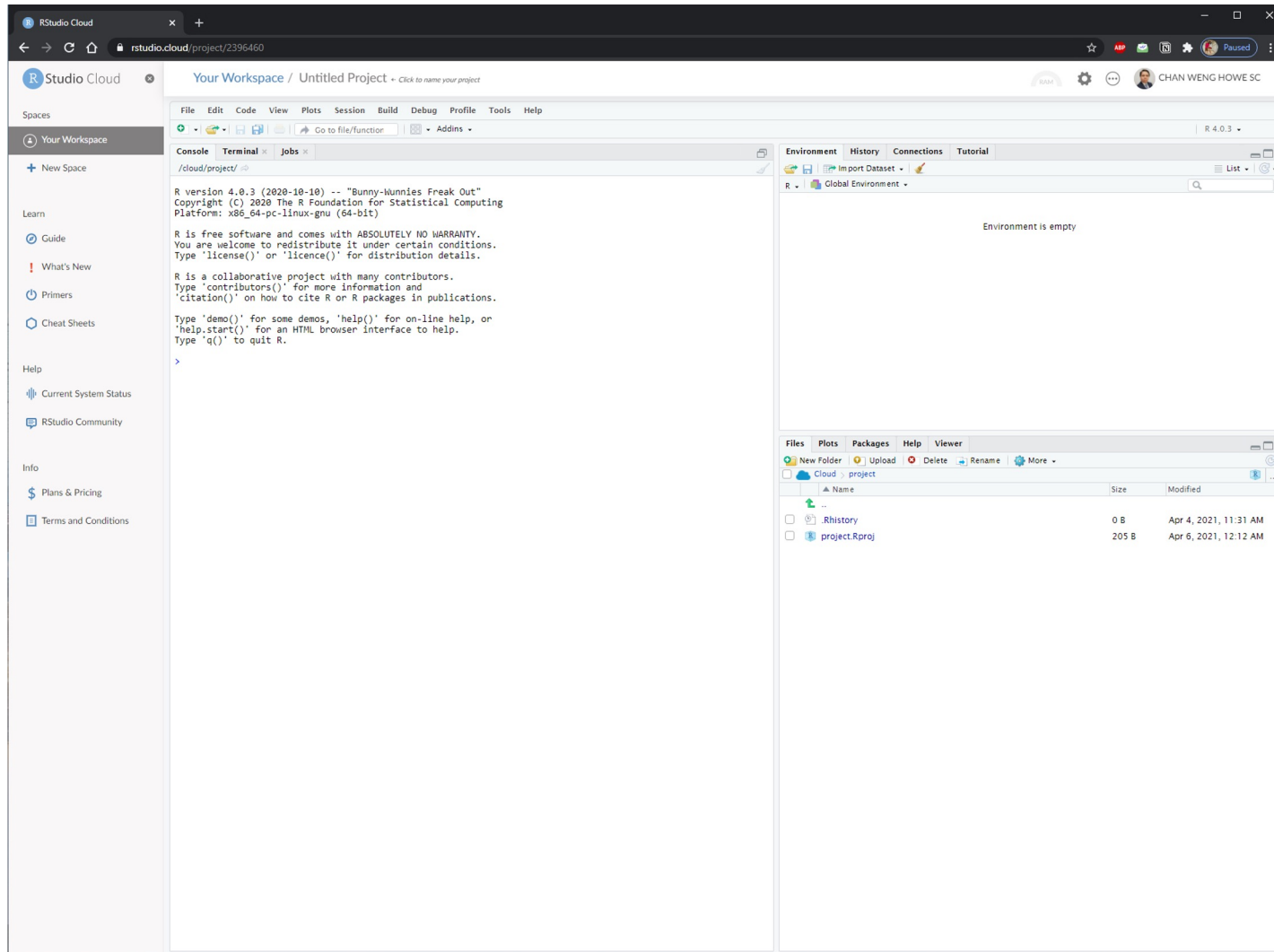
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- AVAILABLE PRICING PLANS
- RSTUDIO CLOUD GUIDE
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Using RStudio Cloud



Setting up R in local computer

- 2 components:
 - R Programming environment (<https://www.r-project.org/>)
 - RStudio IDE for R
(<https://www.rstudio.com/products/rstudio>)
- R Programming environment provides the **core files** and **libraries** of R
- RStudio act as the Integrated Development Environment (IDE) that provides an **additional layer of GUI** which is consistent across Windows, macOS, Linux and RStudio Cloud.

Installation of R 4.0.5

- R 4.0.5 can be downloaded at:
 - <https://cran.r-project.org/bin/macosx/R-4.0.5.pkg> (macOS)
 - <https://cran.r-project.org/bin/windows/base/R-4.0.5-win.exe> (Windows)
- This should be installed before the installation of RStudio.

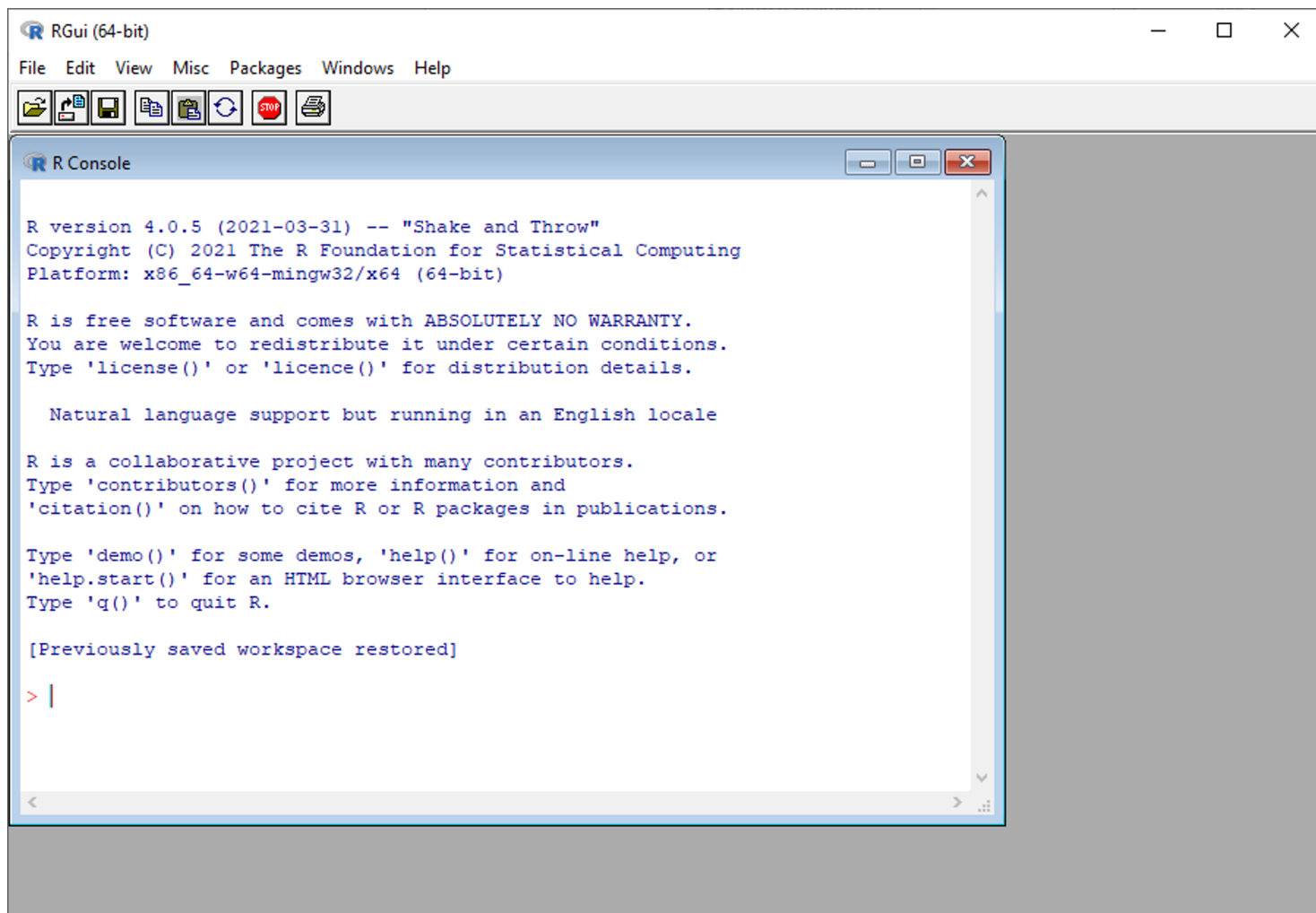
Installation of Rstudio IDE

- RStudio can be downloaded at:
 - <https://download1.rstudio.org/desktop/macos/RStudio-1.4.1106.dmg> (macOS)
 - <https://download1.rstudio.org/desktop/windows/RStudio-1.4.1106.exe> (Windows)

Accessing R in local computer

- 3 Ways to access R in Windows / macOS:
 - i. via **R Console** (come together with R environment)
 - ii. via **Command Prompt** (Win) / **Terminal** (macOS)
 - Type “R” in the command prompt or terminal and press enter to initialize the R environment.
 - iii. via **RStudio**
 - A more organized and integrated UI

R Console Interface



The screenshot shows the RGui (64-bit) application window. The title bar reads "RGui (64-bit)". The menu bar includes "File", "Edit", "View", "Misc", "Packages", "Windows", and "Help". Below the menu bar is a toolbar with icons for file operations and execution. The main window contains an "R Console" pane with the following text:

```
R version 4.0.5 (2021-03-31) -- "Shake and Throw"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

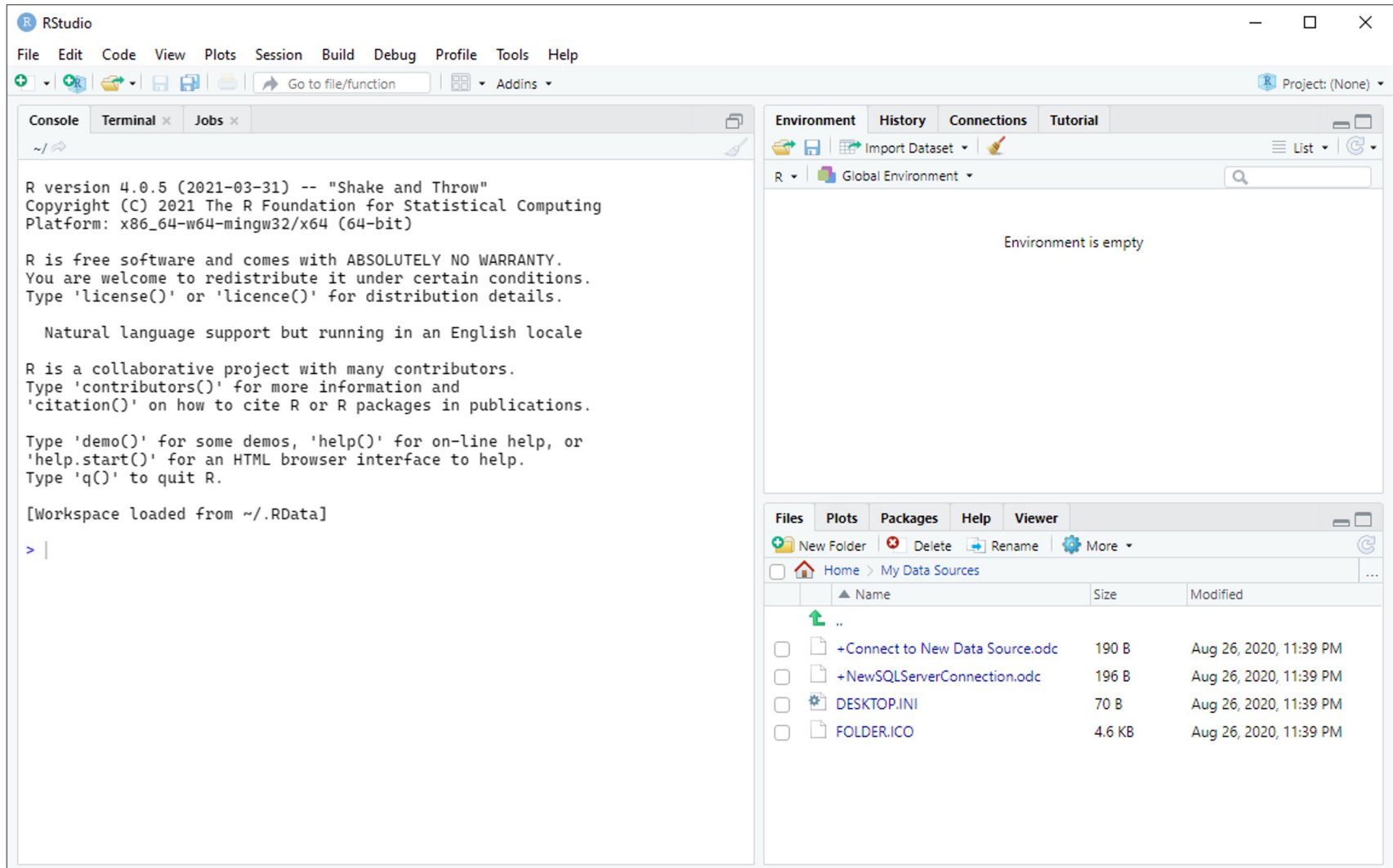
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

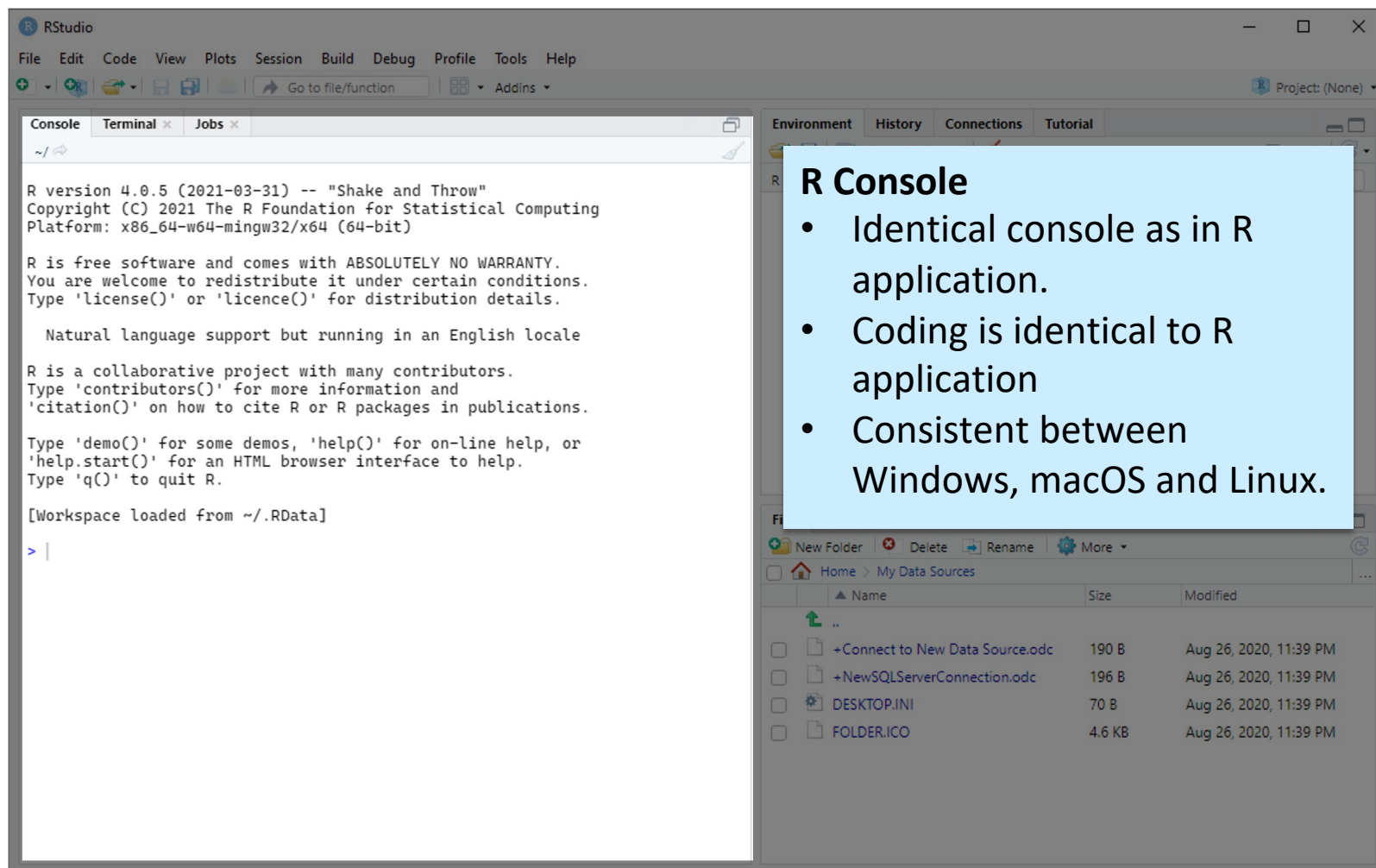
[Previously saved workspace restored]

> |
```

RStudio Interface



RStudio Interface



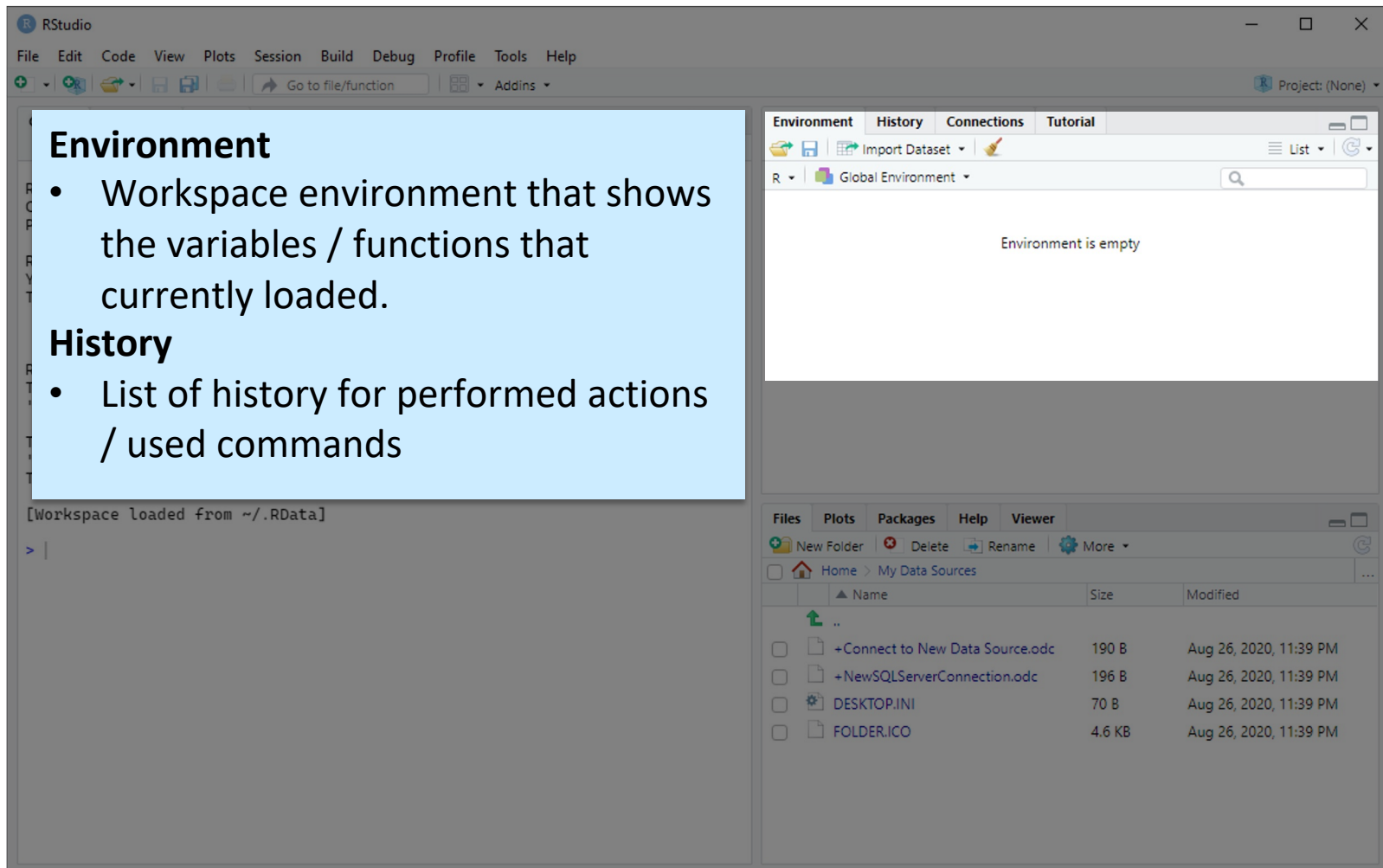
RStudio Interface

Environment

- Workspace environment that shows the variables / functions that currently loaded.

History

- List of history for performed actions / used commands



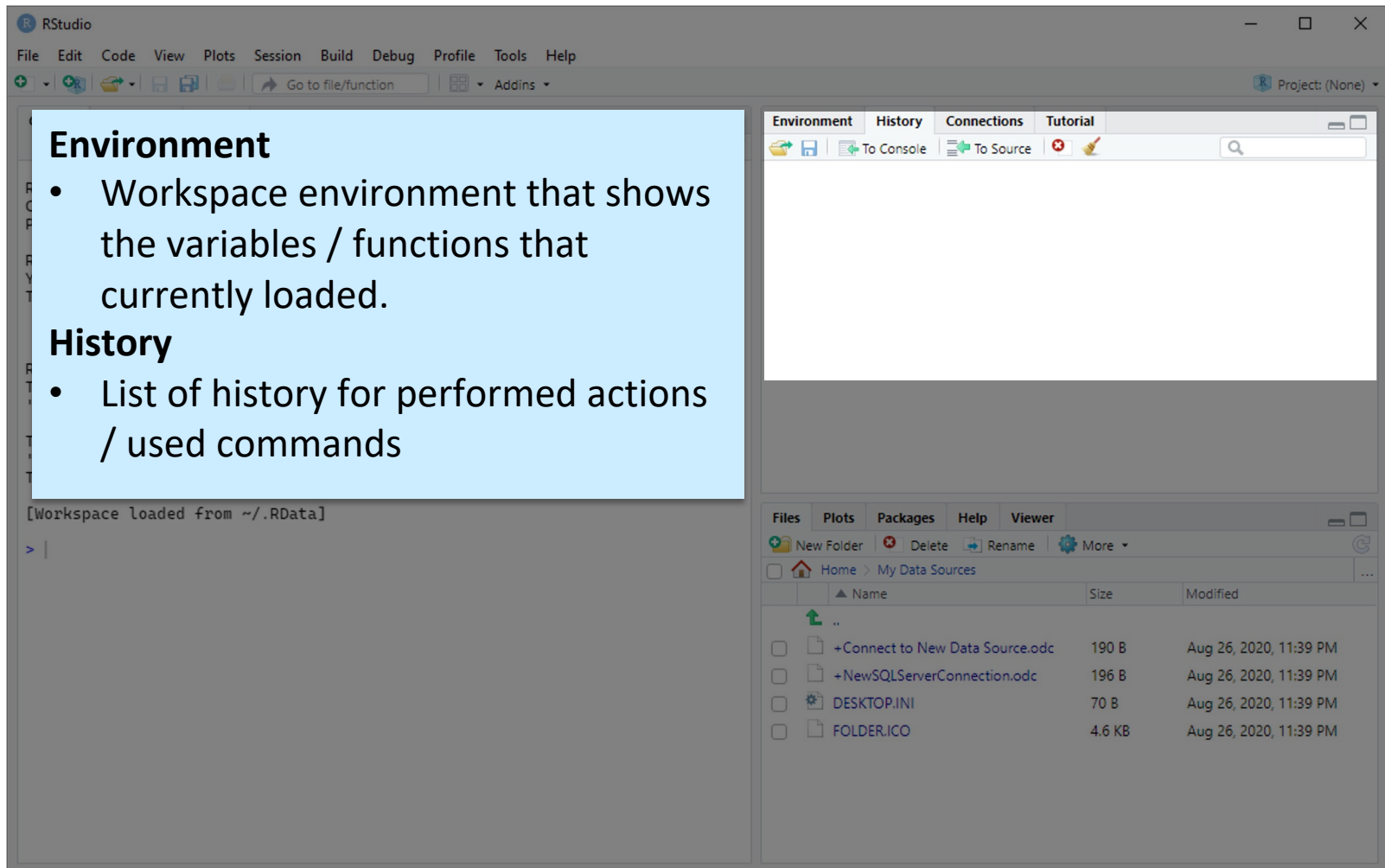
RStudio Interface

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RStudio Interface

Files

- View the folder structure and the contents of the current working directory.

Plots

- Display and settings of the charts / graph

Packages

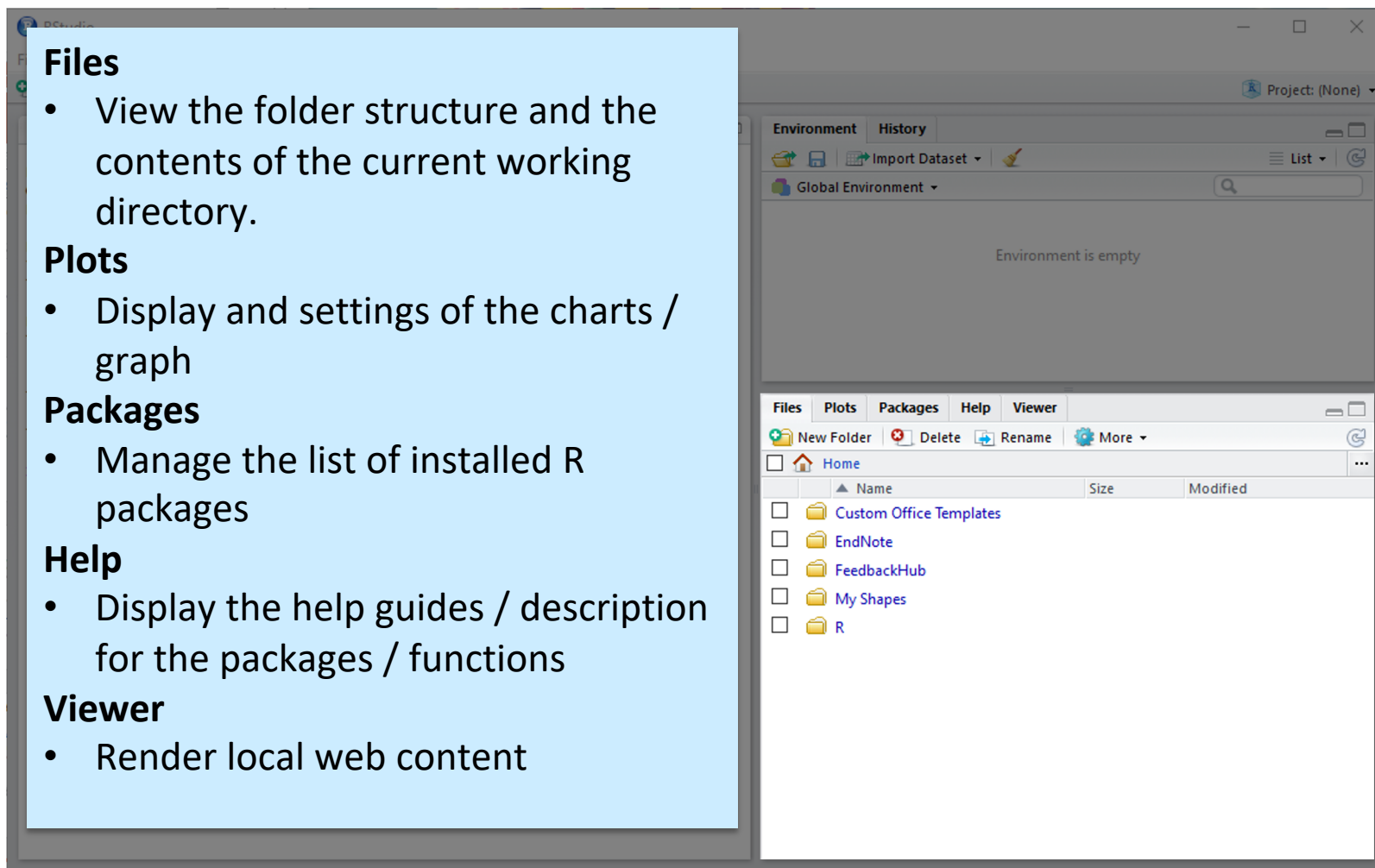
- Manage the list of installed R packages

Help

- Display the help guides / description for the packages / functions

Viewer

- Render local web content



RStudio Interface

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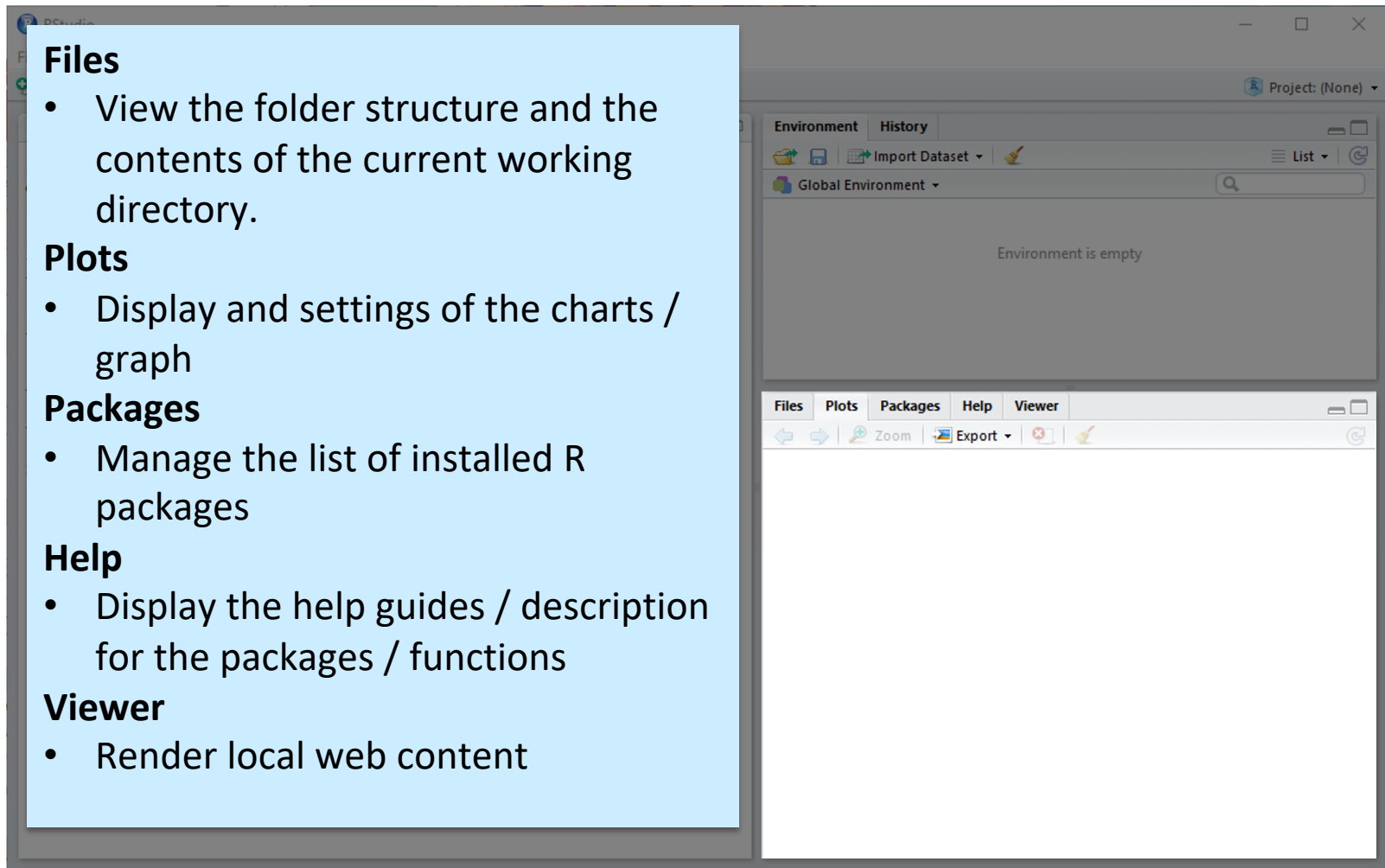
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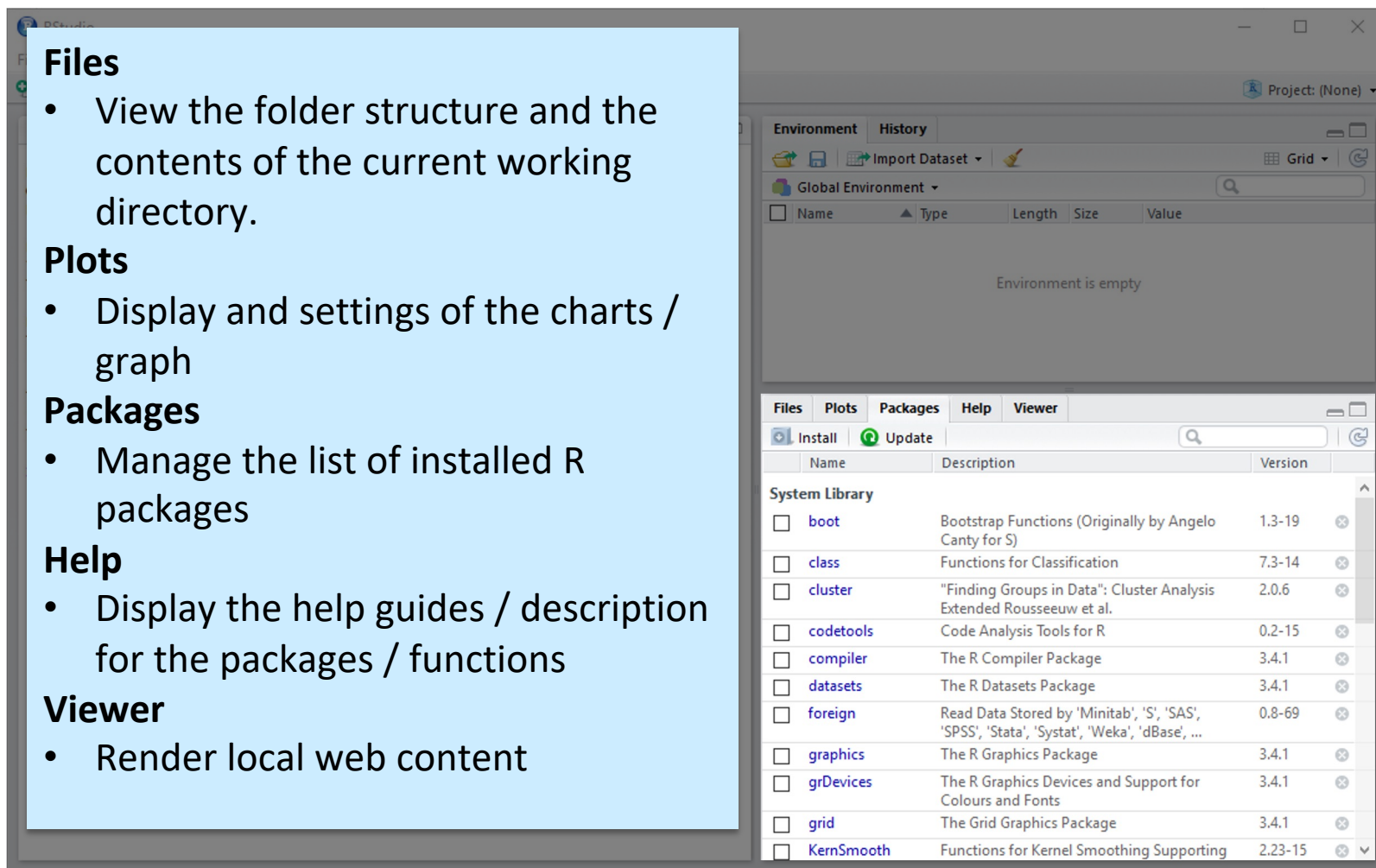
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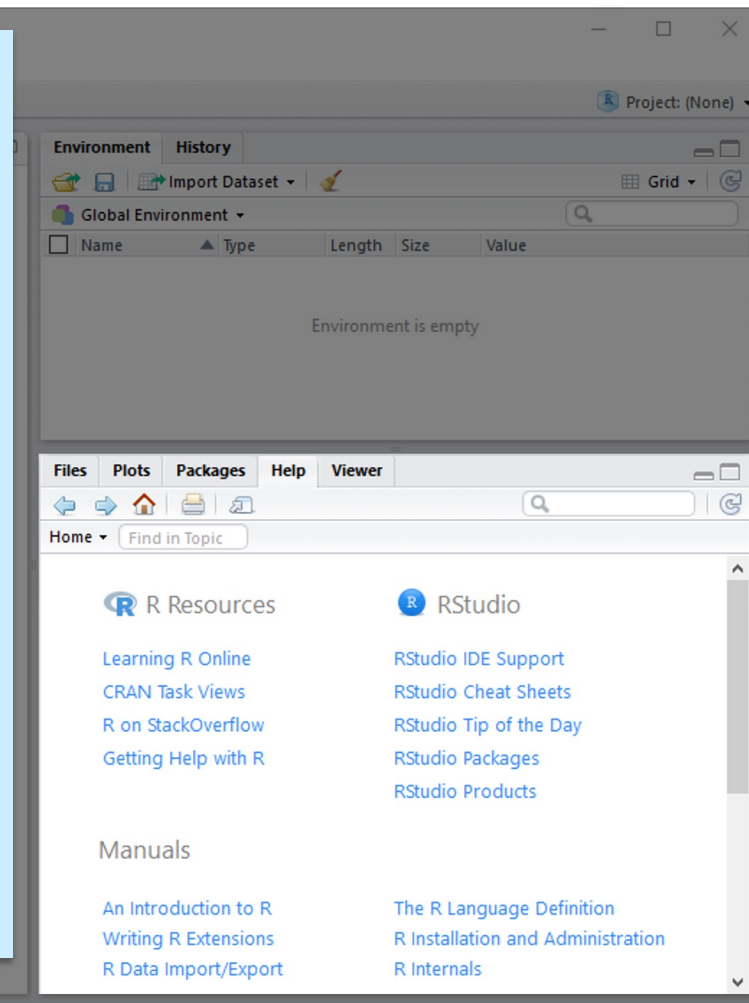
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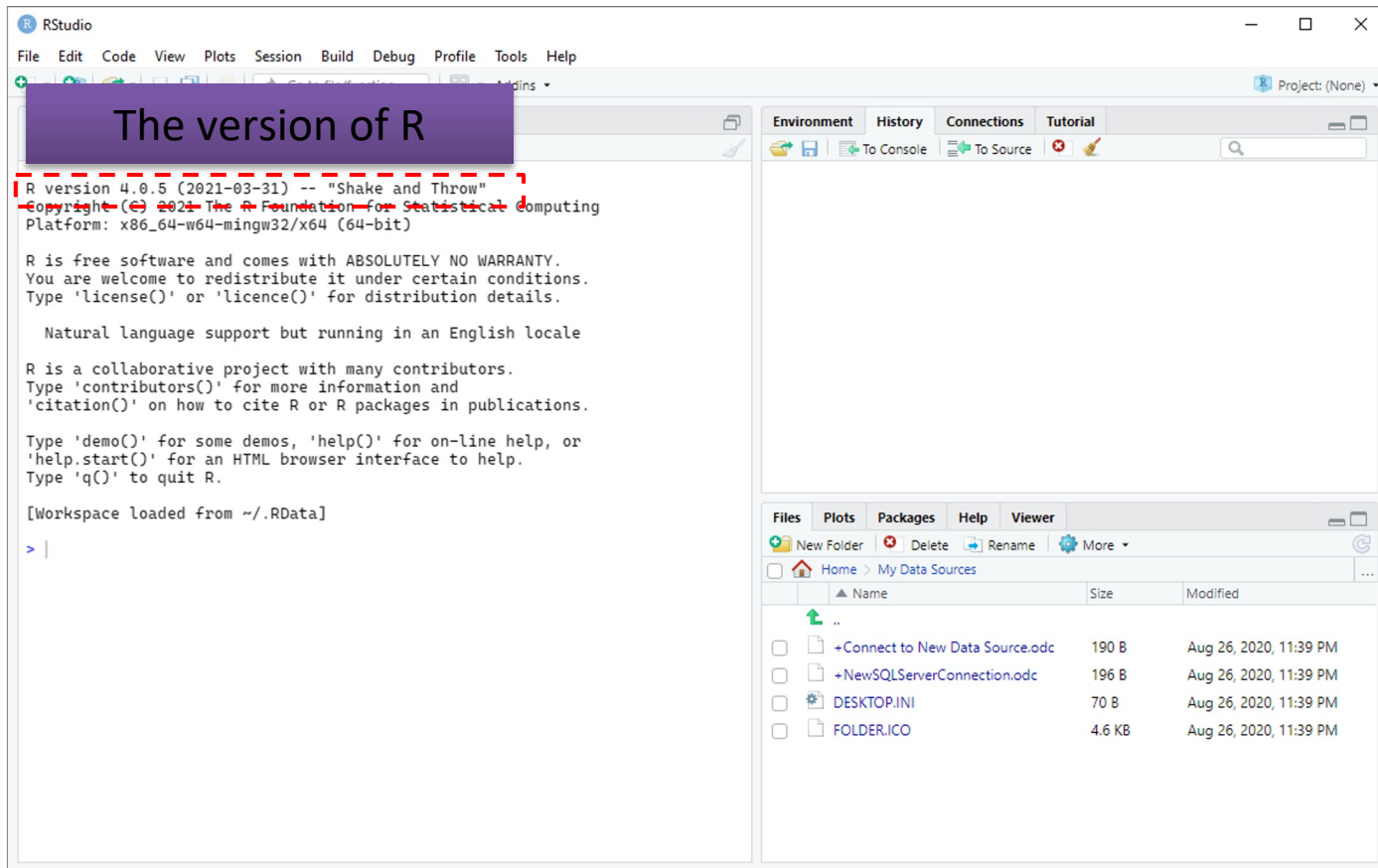
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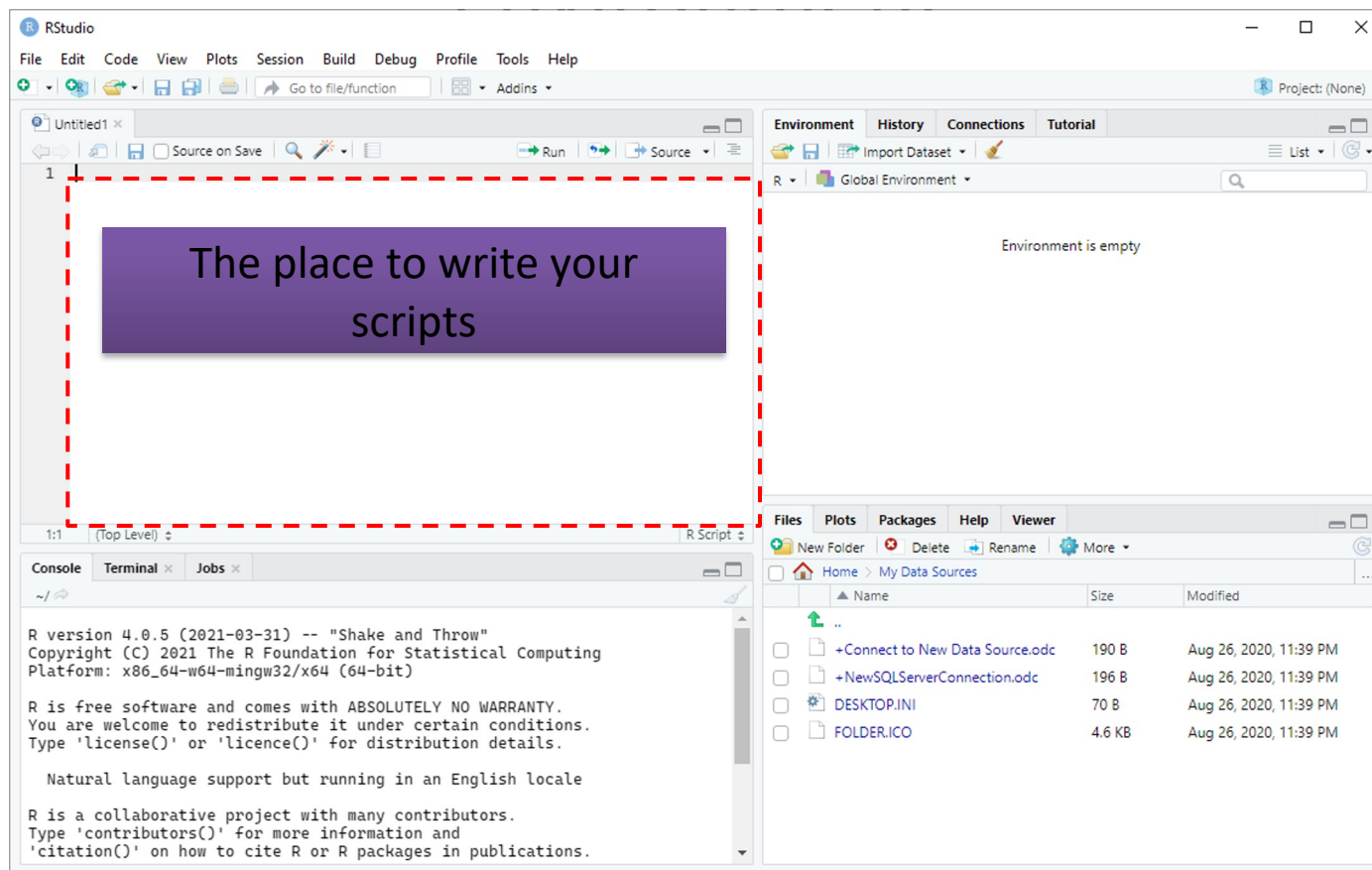
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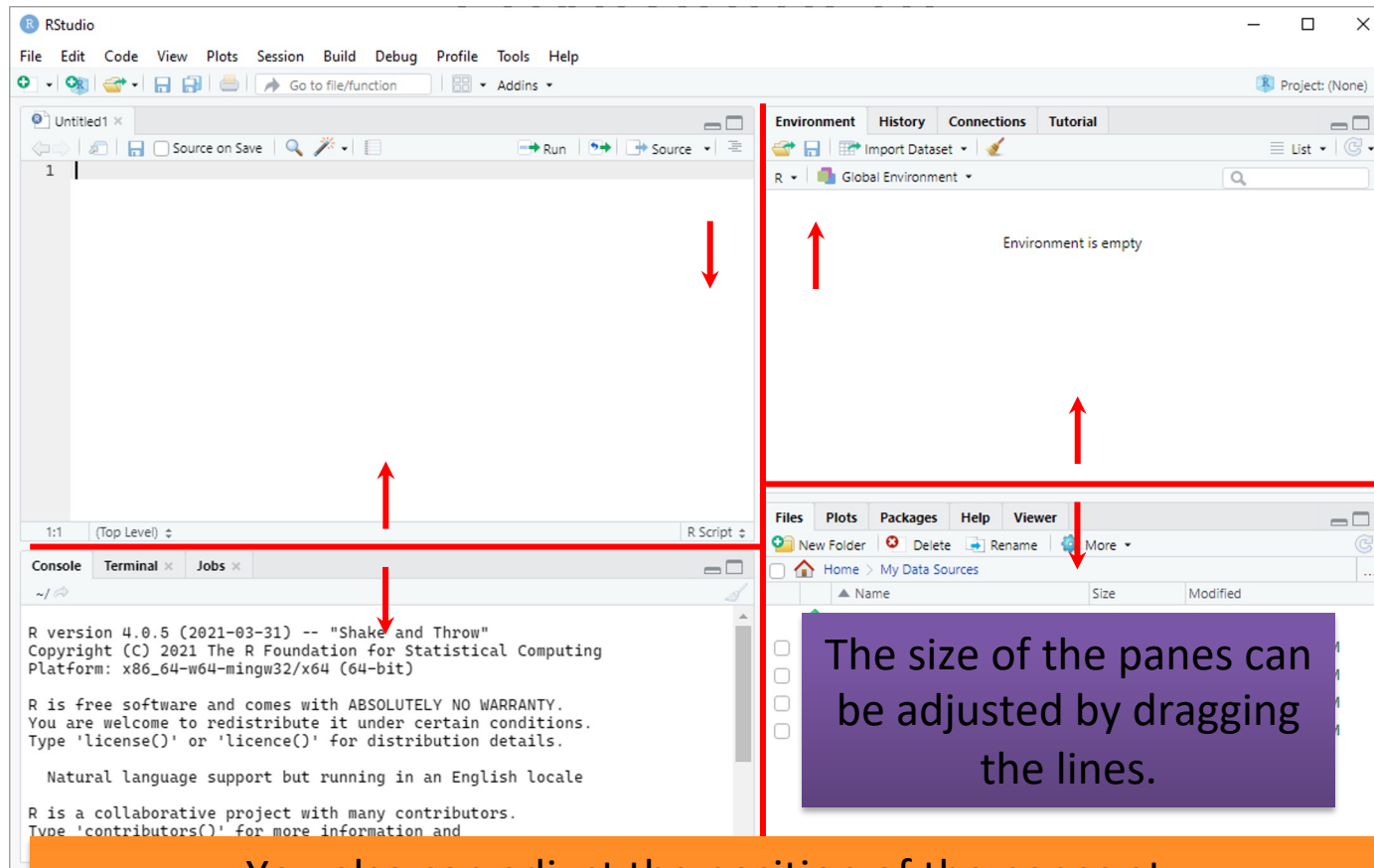
Getting Started with the R Environment



Getting Started with the R Environment



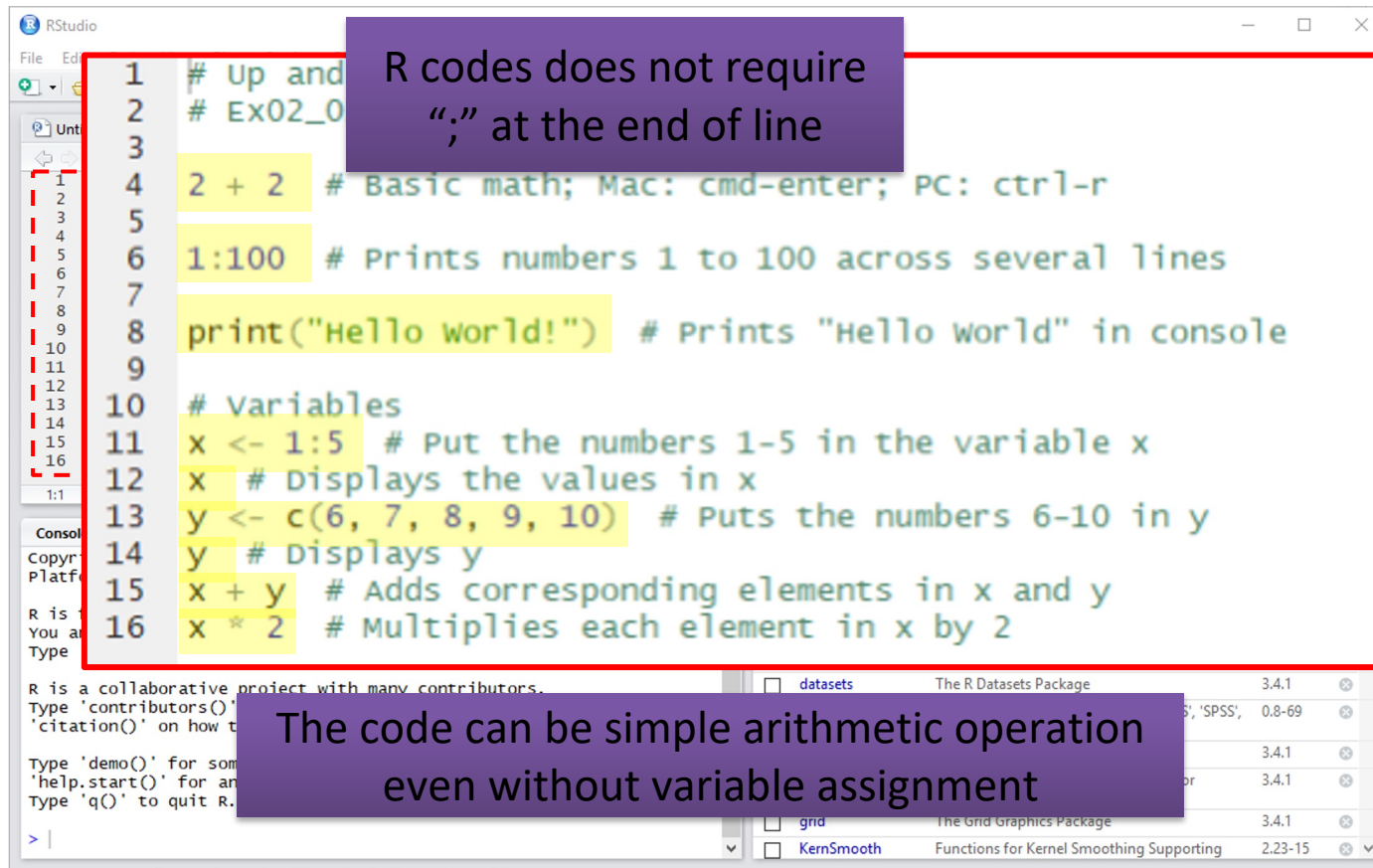
Getting Started with the R Environment



You also can adjust the position of the panes at:
Tools > Global Options > Pane Layout

Getting Started with the R Environment

Getting Started with the R Environment



```
1 # Up and
2 # Ex02_0
3
4 2 + 2 # Basic math; Mac: cmd-enter; PC: ctrl-r
5
6 1:100 # Prints numbers 1 to 100 across several lines
7
8 print("Hello world!") # Prints "Hello world" in console
9
10 # Variables
11 x <- 1:5 # Put the numbers 1-5 in the variable x
12 x # Displays the values in x
13 y <- c(6, 7, 8, 9, 10) # Puts the numbers 6-10 in y
14 y # Displays y
15 x + y # Adds corresponding elements in x and y
16 x * 2 # Multiplies each element in x by 2
```

R codes does not require
";" at the end of line

The code can be simple arithmetic operation
even without variable assignment

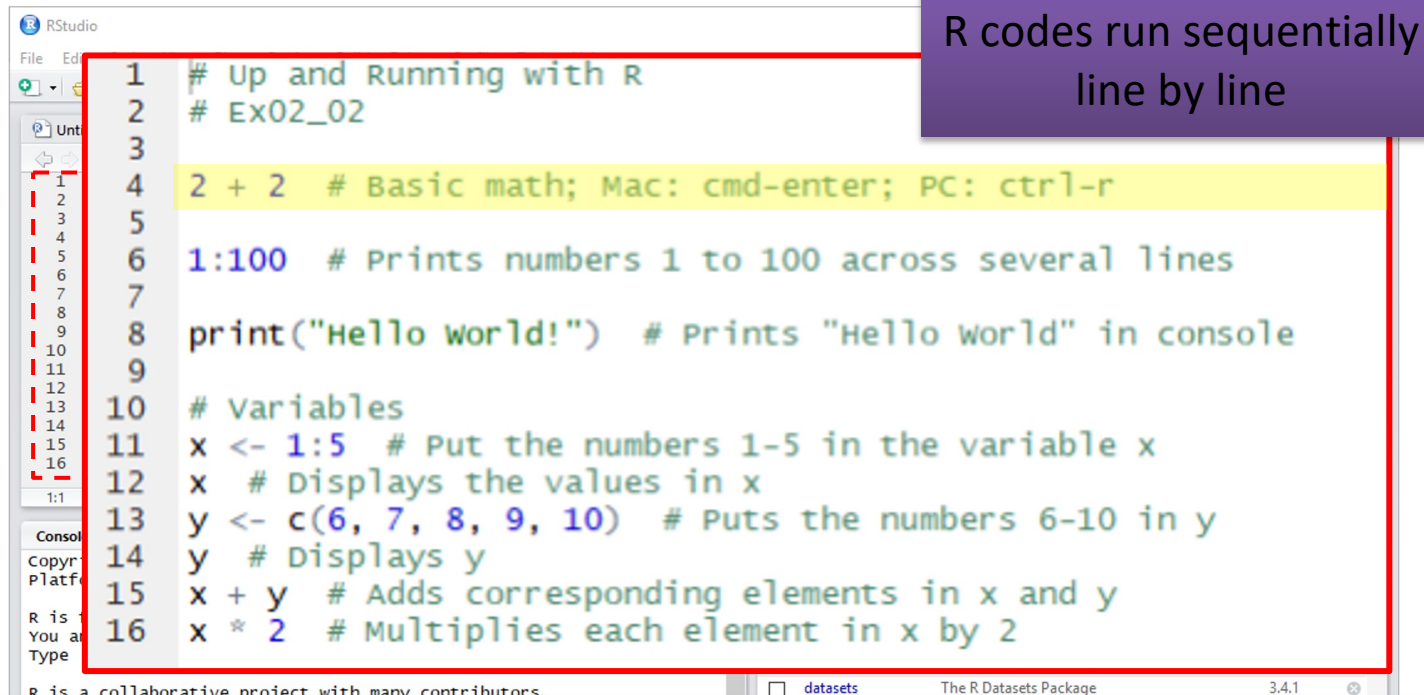
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> |

Package	Version	Source
datasets	3.4.1	The R Datasets Package
grid	3.4.1	The Grid Graphics Package
KernSmooth	2.23-15	Functions for Kernel Smoothing Supporting

Getting Started with the R Environment



```
1 # Up and Running with R
2 # Ex02_02
3
4 2 + 2 # Basic math; Mac: cmd-enter; PC: ctrl-r
5
6 1:100 # Prints numbers 1 to 100 across several lines
7
8 print("Hello world!") # Prints "Hello world" in console
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10 # Variables
11 x <- 1:5 # Put the numbers 1-5 in the variable x
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```

R codes run sequentially
line by line

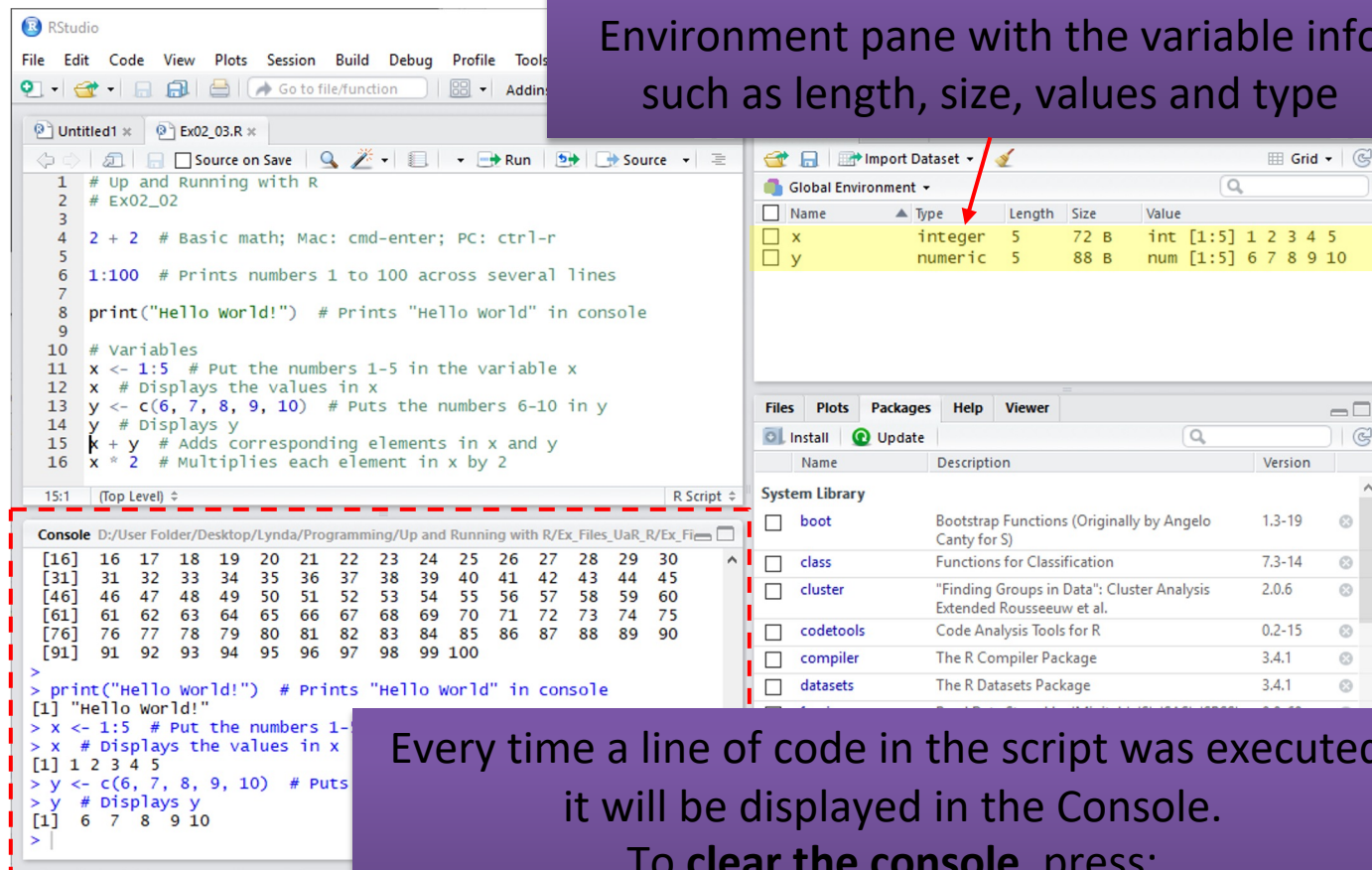
You can run a single line of code by putting a cursor at the start of the line, or run specific part of code by highlighting the code,

Then, press:

Command + **Enter** (macOS) / **CTRL** + **Enter** (Windows)

Getting Started with the R Environment

Variables created will be listed in the Environment pane with the variable info such as length, size, values and type



The screenshot shows the RStudio interface with three main panes:

- Script Editor:** Contains R code for a simple program.


```

1 # Up and Running with R
2 # Ex02_02
3
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14 y # Displays y
15 x + y # Adds corresponding elements in x and y
16 x * 2 # Multiplies each element in x by 2
      
```
- Environment Pane:** Lists variables created in the session.

Name	Type	Length	Size	Value
x	integer	5	72 B	int [1:5] 1 2 3 4 5
y	numeric	5	88 B	num [1:5] 6 7 8 9 10
- Console:** Shows the output of the executed code.


```

[16] 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
[31] 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45
[46] 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
[61] 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75
[76] 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90
[91] 91 92 93 94 95 96 97 98 99 100

> print("Hello world!") # Prints "Hello world" in console
[1] "Hello world!"

> x <- 1:5 # Put the numbers 1-
> x # Displays the values in x
[1] 1 2 3 4 5

> y <- c(6, 7, 8, 9, 10) # Puts
> y # Displays y
[1] 6 7 8 9 10

>
      
```

Every time a line of code in the script was executed, it will be displayed in the Console.

To clear the console, press:

Command + **L** (macOS) / **CTRL** + **L** (Windows)